8051 Microcontroller

I/O Ports, Reset, and Timing (Lecture 8)

1. I/O Ports

- The 8051 has 40 pins.
- 32 pins are used as I/O pins.
- Each port is 8 pins wide, so 32 / 8 = 4 ports.
- The 8051 therefore has 4 I/O ports (or 4 × 8 = 32 I/O pins).

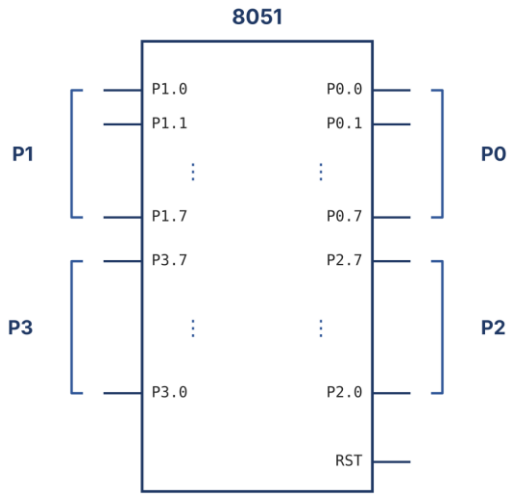


Figure 1. The four I/O ports of the 8051 (simplified; the pin order follows the lecture sketch).

All ports (I/O pins) can be used as input / output.

1.1 Port 0 (P0)

- It must be connected to pull-up resistors (10 kΩ).
- It is used to connect the 8051 with external memory.

  It carries the low part of the addresses: P0 ↔ AD0 – AD7.

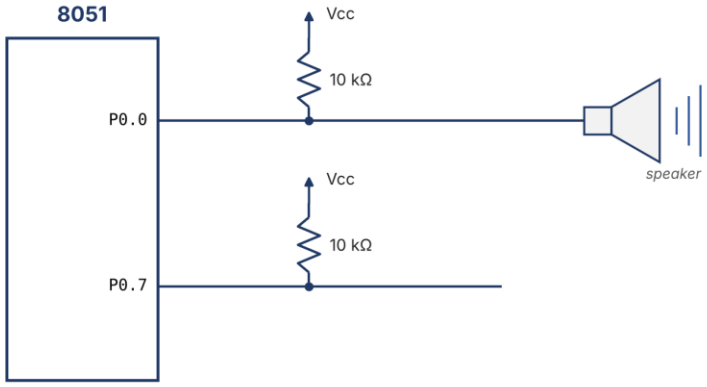


Figure 2. P0 pins with 10 kΩ pull-up resistors to Vcc; P0.0 drives a speaker.

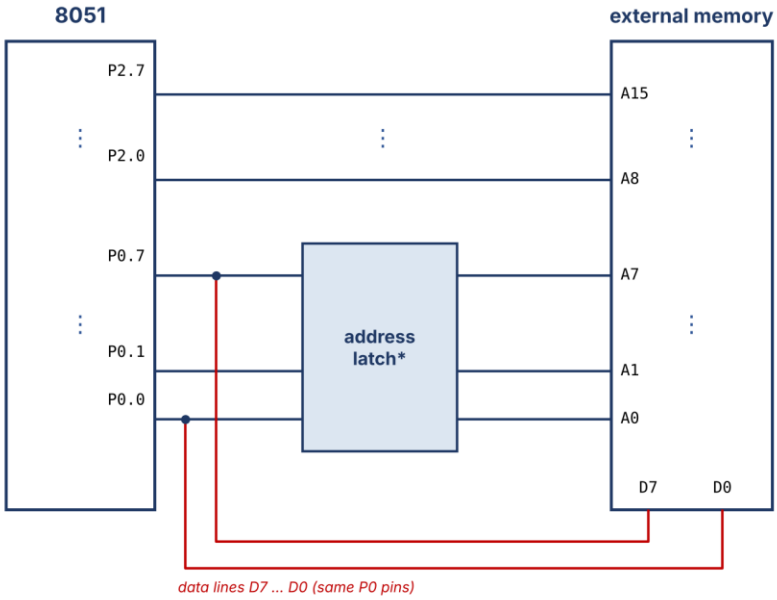


Figure 3. Connecting external memory: P2 supplies A8–A15, P0 supplies A0–A7 (through a latch) and the data lines.

** The box between P0 and the low address pins is the address latch that holds the low address byte while P0 is reused for data.*

1.2 Port 1 (P1)

- It is used only as input / output (no additional task).

1.3 Port 2 (P2)

- It is used to connect external memory.
- It carries the high part of the addresses (A8 – A15).

1.4 Port 3 (P3)

- Each pin has a special task (see slide #7).
- P3.6 → WR (write signal, active low) } external memory

- P3.7 → RD (read signal, active low) } external memory
- Other control pins of the external-memory interface: EA, PSEN, ALE (see slide #6).

Port	Besides I/O, it is used for	Notes
P0	External memory: low address byte (AD0–AD7)	Needs 10 kΩ pull-ups
P1	Nothing, I/O only	
P2	External memory: high address byte (A8–A15)	
P3	Special task for each pin (e.g., P3.6 = WR, P3.7 = RD)	See slide #7

2. The Reset Pin (RST)

RST is the reset pin. It is used to terminate all activities and start with initial values (upon reset). The initial values are:

Register	Value after reset
A, B, R0 → R7	00H
PC, DPTR	0000H
SP	07H
P0, P1, P2, P3	FFH

3. Configuring a Port as Input or Output

- By default, all ports are input ports (they hold FFH after reset).
- The programmer can make any I/O pin an input or an output.
- **To make any pin an input, send 1 to it.**
- **To make any pin an output, send 0 to it.**

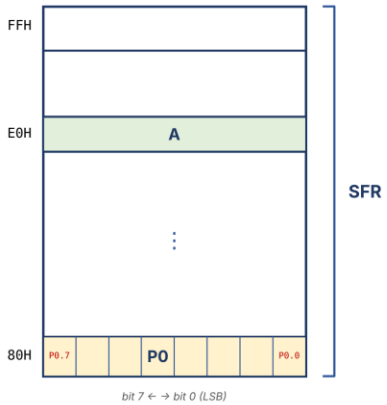


Figure 4. The ports and the accumulator live in the SFR area: A is at E0H and P0 is at 80H (P0.0 is the least significant bit).

3.1 Examples

Configure P3 as input:

```
        MOV   P3, #11111111B
        MOV   P3, #0FFH          ; same thing in hex
```

Configure single pins:

```
        SETB  P1.0               ; P1.0 is an input pin
        CLR   P1.2               ; P1.2 is an output pin
```

Configure P2 to be an output port:

```
        MOV   P2, #00H      ; or MOV P2,#0
```

4. Controlling an LED from a Port Pin

The same pin can turn an LED on or off, but the logic level that lights the LED depends on how the LED is wired.

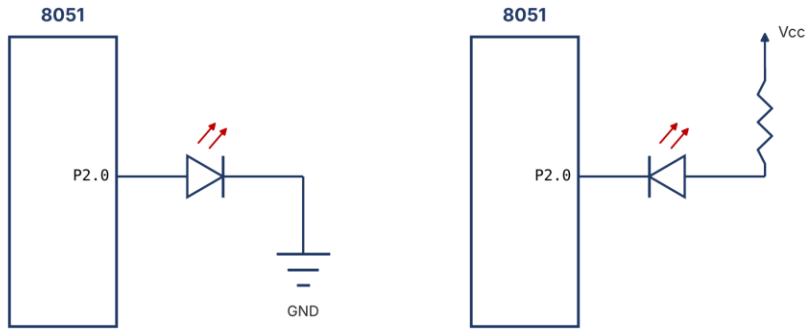


Figure 5. Left: LED between P2.0 and ground (active high). Right: LED between Vcc (through a resistor) and P2.0 (active low).

Circuit	Turn the LED ON	Turn the LED OFF
LED between P2.0 and Gnd	`SETB P2.0`	`CLR P2.0`
LED between Vcc and P2.0	`CLR P2.0`	`SETB P2.0`

5. Example Programs

5.1 Write a code to read P0 and send its 2's complement to P1 continuously

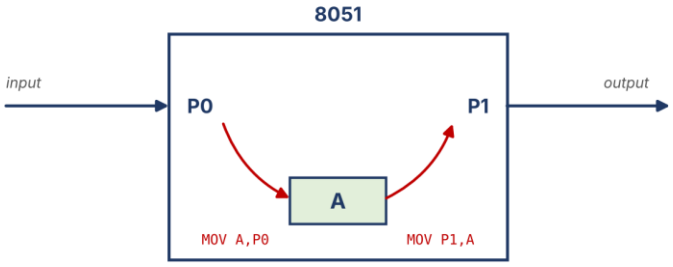


Figure 6. Data path: P0 → A → P1.

```
        ORG   0000H
        MOV   P0, #0FFH      ; P0 as input
Back:   MOV   A, P0
        CPL   A              ; 1's complement
        INC   A              ; +1  ->  2's complement
        MOV   P1, A
        SJMP  Back
```

5.2 Toggle P1 and P2 continuously

Toggling a pin means complementing it (1 becomes 0 and 0 becomes 1).

Solution 1 — using CPL:

```
Back:   MOV   A, P1
        CPL   A
        MOV   P1, A
        MOV   A, P2
        CPL   A
        MOV   P2, A
        ACALL DELAY
        SJMP  Back

DELAY:  MOV   R2, #200
HERE:   DJNZ  R2, HERE
        RET
```

Solution 2 — without CPL:

```
Back:   MOV   A, #55H
        MOV   P1, A
        MOV   P2, A
        ACALL DELAY
        MOV   A, #0AAH
        MOV   P1, A
        MOV   P2, A
        ACALL DELAY
        SJMP  Back
```

Value	Binary (bit 7 → bit 0)	Pins
55H	0 1 0 1 0 1 0 1	0 = off, 1 = on, alternating
AAH	1 0 1 0 1 0 1 0	the opposite pattern (the complement of 55H)

5.3 Write a program to create a square wave with 50% duty cycle on P1.4

Duty cycle = Time of 1 / T (Time 1 = the time the signal is high, T = the period)

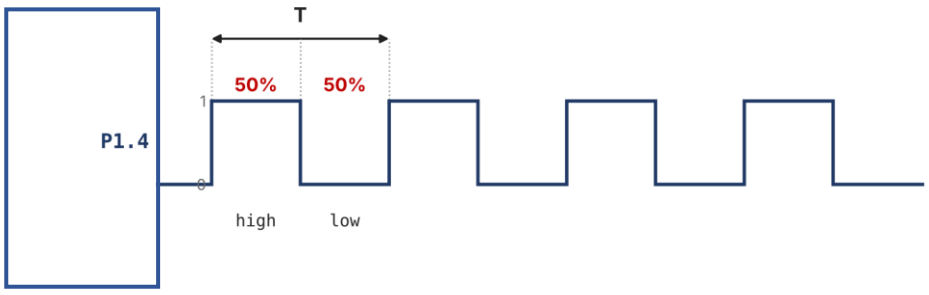


Figure 7. 50% duty cycle: high for half of the period, low for the other half.

```
        ORG   0
Back:   SETB  P1.4
        ACALL DELAY
        CLR   P1.4
        ACALL DELAY
        SJMP  Back

DELAY:  MOV   R4, #250
AGAIN:  NOP
        DJNZ  R4, AGAIN
        RET
```

NOP: No operation (do nothing). It only updates the PC.

5.4 Write a code to create a wave with 75% duty cycle on P2.7

Let **x** be the time the pin is high and **y** the time it is low (one DELAY). Then x / y = 75 / 25 = 3, so x = 3y: keep the pin high for three delays and low for one.

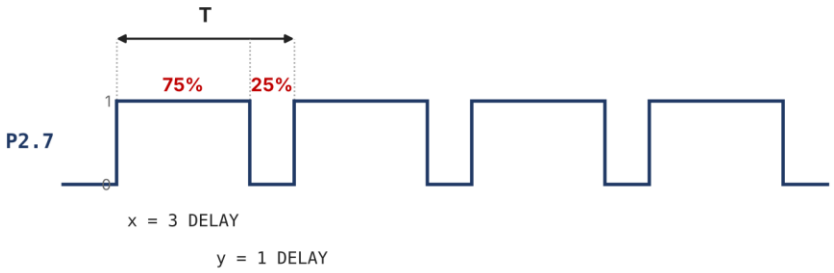


Figure 8. 75% duty cycle: high for three DELAY periods (x), low for one (y).

```
        ORG   0
Back:   SETB  P2.7
        ACALL DELAY
        ACALL DELAY
        ACALL DELAY
        CLR   P2.7
        ACALL DELAY
        SJMP  Back
```

The DELAY subroutine is not repeated in the notes; use the one from the previous example.

Another case — 66% duty cycle: x / y = 66 / 34 ≈ 2, so keep the pin high for about two delays and low for one.

6. Clock, Machine Cycle, and Delay Time

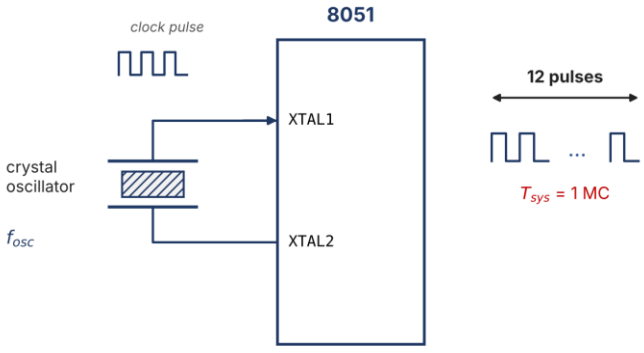


Figure 9. The crystal oscillator drives XTAL1/XTAL2; the 8051 counts 12 clock pulses per machine cycle.

- The 8051 divides the crystal frequency by 12:

$F_{sys} = f_{osc} / 12$ or $f_{sys} = f_{XTAL} / 12$

- The machine cycle (MC) is the system period:

$MC = 12 / f_{osc}$ => $T_{sys} = MC = 12 / f_{osc}$

6.1 Example: f_{osc} = 11.0592 MHz

```
f_sys = 11.0592 MHz / 12 = 0.9216 MHz
MC    = 12 / 11.0592 MHz = 1.085 µs
```

6.2 How long does an instruction take?

Instructions take 1, 2, or 4 machine cycles:

Instruction length	Oscillator cycles
1 MC	12 cycles
2 MC	24 cycles
4 MC	4 × 12 = 48 cycles

6.3 Example: execution time of a code fragment (24 MHz crystal)

Assume the crystal oscillator is 24 MHz. Calculate the time of the following code.

Instruction	MC
MOV A, R4	1
NOP	1
NOP	1
MUL AB	4
MOV R2, B	2

```
MC   = 12 / f_osc = 12 / 24 MHz = 0.5 µs
Time = (1 + 1 + 1 + 4 + 2) x MC
     = 9 x 0.5 µs = 4.5 µs
```

6.4 Example: delay of a loop (6 MHz crystal)

Assume XTAL = 6 MHz. Compute the time delay of this subroutine.

Instruction	MC	Times executed	MC total
DELAY: MOV R2, #100	1	1	1 × 1 = 1

Instruction	MC	Times executed	MC total
NOP	1	1	1 × 1 = 1
HERE: DJNZ R2, HERE	2	100	2 × 100 = 200
RET	2	1	2 × 1 = 2

```
MC         = 12 / 6 MHz = 2 µs
Time delay = (1 + 1 + 200 + 2) x MC
           = 204 x 2 µs = 408 µs
```

6.5 Homework #4 — Question 2 (as far as it appears in the notes)

Given f_osc = 12 MHz, the machine cycle is:

```
MC = 12 / f_osc = 12 / 12 MHz = 1 µs
```

The delay subroutine for this question is the nested loop below. The handwritten solution leaves the final sum blank; it is completed here.

```
DELAY:  CLR   A
        MOV   R2, #4
ME2:    MOV   R1, #50H
ME:     MOV   P1, A
        CLR   A
        CLR   A
        NOP
        DJNZ  R1, ME
        DJNZ  R2, ME2
        RET
```

The inner loop runs 50H = 0101 0000B = 64 + 16 = 80 times, and the outer loop runs 4 times, so the inner body is executed 80 × 4 = 320 times.

Instruction	MC	Times executed	MC total
CLR A	1	1	1
MOV R2, #4	1	1	1
MOV R1, #50H	1	4	4
MOV P1, A	1	320	320
CLR A	1	320	320
CLR A	1	320	320
NOP	1	320	320

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Instruction	MC	Times executed	MC total
DJNZ R1, ME	2	320	640
DJNZ R2, ME2	2	4	8
RET	2	1	2
Total			**1936**

```
Delay = (1 + 1 + 4 + 320 + 320 + 320 + 320 + 640 + 8 + 2) x MC
      = 1936 x 1 µs = 1936 µs  (about 1.94 ms)
```

Proofreading Notes

Spelling, grammar, and punctuation were corrected throughout. The technical content was kept as written, except where flagged below.

- **Missing pages:** the handwritten pages are numbered 1–9 and then 12. Pages 10 and 11 are not in the scan (they probably cover instruction timing). The last page of the PDF is blank.
- **Slide references:** slide #6 (EA, PSEN, ALE) and slide #7 (special functions of P3) are referenced in the notes but their content is not included.
- **Toggle program (Solution 2):** no delay after the AAH write; see the note in Section 5.2.
- **Timing example (24 MHz):** MOV R2,B is counted as 1 MC but takes 2 MC; see the note in Section 6.3.
- **Homework #4:** only Question 2 appears in the scan, and its final sum was blank (completed in Section 6.5: 1936 µs).
- **LED sketch:** the left-hand circuit has no series resistor. A stray sketch near the top of that page was not transcribed.
- **Figure 3:** the box between P0 and the address pins was unlabeled; it is labeled “address latch” here.

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